

Timer-Controlled DOORBELL

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Repeated or prolonged pressing of a doorbell can be disruptive, particularly in homes or offices. Visitors often press the bell multiple times in quick succession or hold it down too long, creating unnecessary disturbance. To counter this, a timer-controlled doorbell device has been developed using two

555 timer ICs in monostable mode, combined with a relay driver and a control circuit to block repeated activations within a short interval.

This compact and effective system introduces a lockout period via IC2, ensuring the bell sounds only once per trigger cycle, regardless of how often the button is pressed. The timing intervals for IC1 and IC2 can be adjusted using appropriate resistor and capacitor values.

Designed with standard, low-cost components, this dual 555 timer-based device offers a simple yet intelligent solution to prevent repeated bell ringing. It adds smart control to a basic doorbell system, improving both convenience and user experience. Fig. 1 shows the prototype on a breadboard.

Circuit and working

Fig. 2 shows the circuit diagram of the timer-controlled doorbell. The device is built using two 555 timer ICs (IC1 and IC2), both configured in monostable mode. A relay (RL1) is used to activate the doorbell, along

with a pair of transistors—BC547 (T1) and 2N3904 (T2). These transistors drive the relay and control the reset functionality of the first timer, respectively. The circuit handles multiple presses of the doorbell button, ensuring that the bell rings only once within a preset interval, regardless of how many times the button is pressed during that period.

The first monostable circuit, based on IC1, is responsible for triggering the actual ringing of the doorbell. When push-button switch S1 is pressed momentarily, it triggers IC1, which outputs a high pulse for a duration determined by resistor R3 and capacitor C2. This output is fed to the base of a BC547 transistor, which functions as a relay driver. When activated, the transistor energises the relay coil, closing its contacts and powering the doorbell connected to the relay. The doorbell sounds for around 5 seconds.

This arrangement ensures that the bell rings for a fixed, short duration—even if the button is held down for an extended period. Once

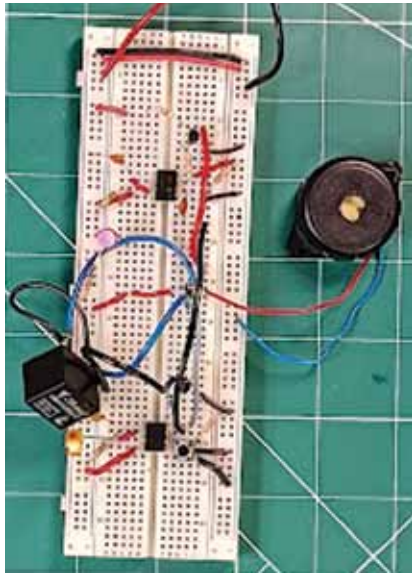


Fig. 1: Author's prototype

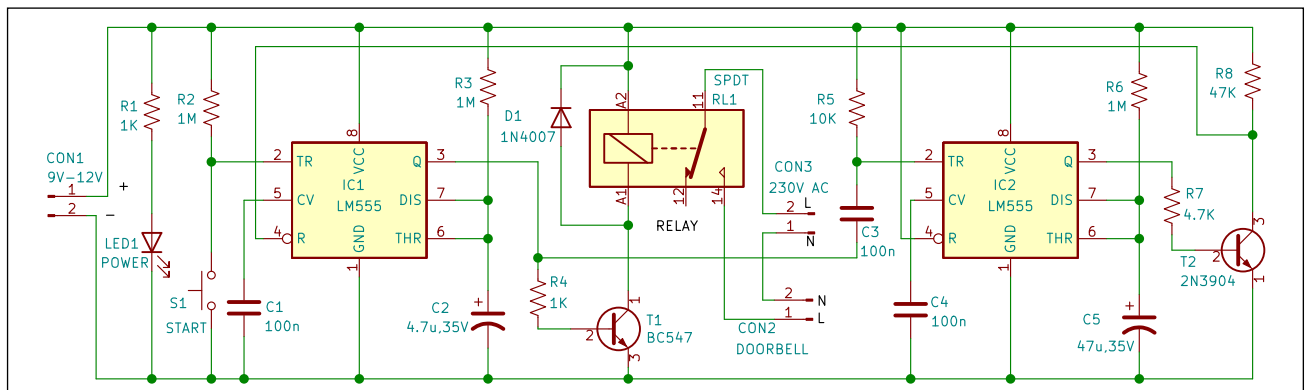


Fig. 2: Circuit diagram